

Problem

A linear network has a current input $7.5 \cos(10t + 30^\circ)$ A and a voltage output $120 \cos(10t + 75^\circ)$ V. Determine the associated impedance.

Step-by-step solution

Step 1 of 5

Draw the following circuit as shown in figure 1.

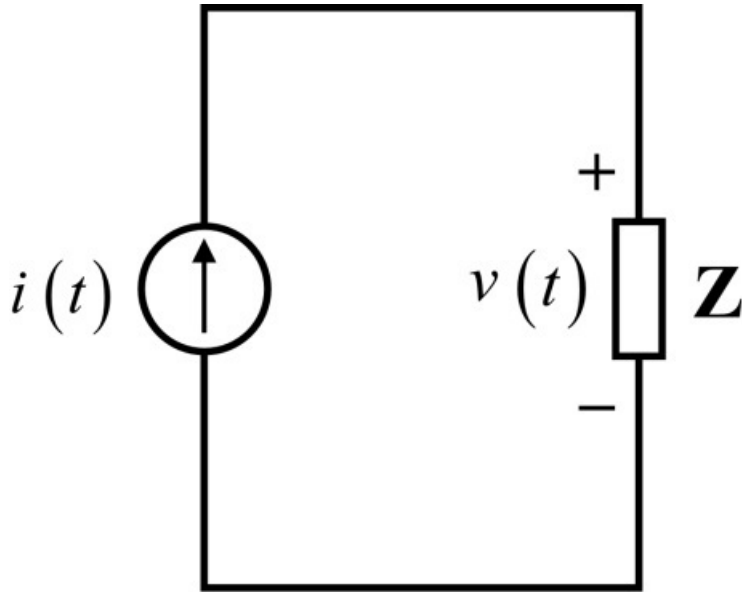


Figure 1

Step 2 of 5

Consider the voltage $\mathbf{V}$ in time domain.

$$v(t) = 120 \cos(10t + 75^\circ) \text{ V}$$

Convert $v(t)$ into phasor form.

$$\mathbf{V} = 120 \angle 75^\circ \text{ V}$$

Step 3 of 5

Consider the voltage $\mathbf{I}$ in time domain.

$$i(t) = 7.5 \cos(10t + 30^\circ) \text{ A}$$

Convert $v(t)$ into phasor form.

$$\mathbf{I} = 7.5 \angle 30^\circ \text{ V}$$

Step 4 of 5

Apply Ohm's law to the circuit shown in figure 1.

$$\mathbf{V} = \mathbf{IZ}$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

Substitute $120\angle 75^\circ$ for $\mathbf{V}$, and $7.5\angle 30^\circ$ for $\mathbf{I}$.

$$\begin{aligned}\mathbf{Z} &= \frac{120\angle 75^\circ}{7.5\angle 30^\circ} \\ &= \frac{120}{7.5} \angle (75^\circ - 30^\circ) \\ &= 16\angle (45^\circ) \, \Omega\end{aligned}$$

Step 5 of 5

Convert $\mathbf{Z}$ into rectangular form.

$$\begin{aligned}\mathbf{Z} &= 16\cos(45^\circ) + j16\sin(45^\circ) \\ &= 16\left(\frac{1}{\sqrt{2}}\right) + j\sin\left(\frac{1}{\sqrt{2}}\right) \\ &= (11.3137 + j11.3137) \, \Omega\end{aligned}$$

Therefore, the associated impedance $\mathbf{Z}$ is $\boxed{16\angle 45^\circ \, \Omega \text{ (or)} (11.3137 + j11.3137) \, \Omega}$.